32 · The female reproductive system: every structure and its job

The words of this track, as it says them. Bold brown is the examiner.

The female reproductive system. Ten marks.

Label all the structures of the female reproductive system, and state their functions.

The other half of her ten-mark question, from her Repro Female deck and her quiz keys. Her figure is a front view, numbered one to eight: the vagina, the cervix, the myometrium, the cavity of the uterus, the fallopian tube, the fimbriae, the ovary, and the endometrium. Picture it as you listen.

The hook: make, catch, carry, grow, deliver. The ovaries make the eggs, the fimbriae catch them, the tubes carry them, the uterus grows the baby, and the vagina is the way out.

The system's function: to produce the female gametes, the ova; to secrete the female sex hormones, oestrogen and progesterone; and to support fertilisation, pregnancy and birth.

The ovaries: paired, one on each side of the uterus, held by ligaments, and supplied by the ovarian artery. The follicles sit in the outer part, the ovarian cortex. Each follicle is an oocyte, an immature egg, surrounded by granulosa cells. The ovaries produce the eggs, and secrete oestrogen and progesterone.

The fallopian tubes, also called the uterine tubes. One end opens into the uterus. The other widens into the ampulla, and ends in a funnel, the infundibulum, fringed with fingerlike fimbriae. The tube doesn't actually touch the ovary: the ovulated oocyte lands in the peritoneal cavity, and the fimbriae draw it in. The tube's jobs: to receive the oocyte, to be the site of fertilisation, and to carry the oocyte toward the uterus, by peristalsis and by its cilia, which help the sperm along too.

The uterus: a hollow, thick-walled organ in the pelvis, behind the bladder and in front of the rectum. Its parts are the fundus, the rounded top above the tubes; the body; the isthmus, the narrow part; and the cervix, the neck, which projects into the vagina. Its function: it receives the embryo, and nourishes it until the placenta forms.

The wall of the uterus has three layers. The perimetrium, the thin outer serous layer. The myometrium, the thick middle layer of smooth muscle, which contracts in labour. And the endometrium, the lining, rich in blood vessels and glands. Its stratum functionalis is shed in each period, and its stratum basalis grows a new one.

The cervix: the neck of the uterus, with the cervical canal running through it, from the internal os to the external os. Its glands make mucus that blocks bacteria spreading up from the vagina.

The vagina: a thin-walled tube from the cervix to the outside, with the urethra in front and the rectum behind. It's the passageway for birth and for menstrual flow, and the female organ of

copulation. It's normally acidic, which keeps bacteria in check. Her quiz: antibiotics can kill off those bacteria, and let yeast overgrow.

The external genitalia, the vulva. The mons pubis, the fatty pad over the pubic bone. The labia majora, the hair-covered outer folds, which match the scrotum. The labia minora, the hair-free inner folds. And the clitoris, erectile tissue under a prepuce. Her key: the clitoris is the female match to the glans penis.

The breasts, the mammary glands, are modified sweat glands. They have their own track, with Pania's case.

The traps. The endometrium receives the embryo; the myometrium is the muscle. Fertilisation happens in the fallopian tube, not the uterus. And the oocytes sit in the ovarian cortex, not the medulla.

Once more. The female system, structure by structure.

The ovaries hold the follicles in their cortex, make the oocytes, and secrete oestrogen and progesterone. The fimbriae of the fallopian tubes draw in the ovulated oocyte; the tube is the site of fertilisation, and cilia and peristalsis carry the oocyte to the uterus. The uterus, with its fundus, body, isthmus and cervix, receives and nourishes the embryo. Its wall is the perimetrium, the myometrium of smooth muscle, and the endometrium, whose functionalis is shed and whose basalis regrows. The cervix's mucus blocks bacteria. The vagina is the birth canal and the copulation organ, and it's acidic. The vulva is the mons pubis, the labia majora and minora, and the clitoris.